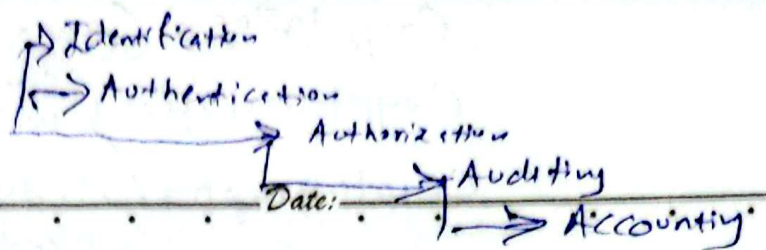
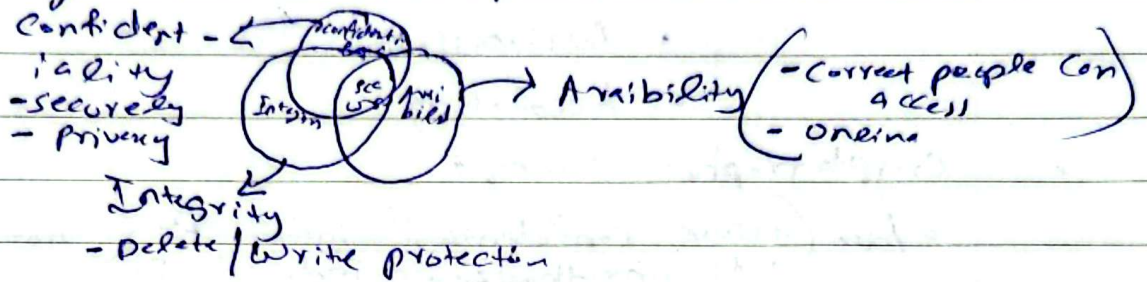


Intro to IS



- SQL Injection (allows an attacker to interfere with the queries an application makes to its database)
- Information security (Task of securing information that is in digital form)
- Three types of information protection: CIA.



a) Confidentiality (For maintaining confidentiality on a network, data must be protected from unauthorized access or v.l.)

- Events that lead to Confidentiality are

- not encrypting before sending it to another person
- falling prey to social engineering attacks
- A classic data breach (Prevention Detection Response)

Info for Systems K:
Accuracy & reliability
is given, and no
changes without
permission

b) Integrity (assurance of accuracy & reliability of information)

- Confidentiality & integrity depends on each other.

- Events that can lead to integrity (non-intentional)

- Accidentally deleting files
- entering invalid data
- altering configuration

c) Availability (authorized subjects are granted timely & access to objects)

Cryptology (area of cryptography & cryptanalysis)
Cryptanalysis (study of techniques for decryption without knowledge of plaintext)
Cryptography (Study of Algorithms used for encryption) Date: _____

- Security Concepts →
 - Confidentiality (authorized party can view information)
 - Integrity (ensures information is not altered) + identity of sender
 - Availability ()
 - Authentication
 - Authorization
 - Accounting
- Cryptographic Concept.
 - Aim (assure confidential information not made available to unauthorized individuals)
 - How (encrypt original data; anyone can see it, but only authorized individuals can decrypt to see original data)
- - Ciphertext (encrypted or coded message)
 - encryption (convert from plaintext to ciphertext)
 - decryption (restore the plaintext from ciphertext)
 - Cipher (A particular Algorithm)
- Cryptography Goals
 - Confidentiality
 - Integrity
 - Authentication
 - Non repudiation (Sender cannot deny having sent a message)
 - ↓
 - provides proof of origin. (often done using digital signatures)

Julius Caesar (send secret military messages)

Date: _____

process of applying that cipher to message

• Cipher & Ciphering

- Cipher is method or technique used to hide real meaning of message.

(a) - Substitution Ciphers: ① Caesar Cipher

↓
each letter in the message is replaced by another letter

② ROT13 (Rotate by 13 places)

- Drawbacks of Caesar And Rot13 (- easily crackable in few seconds)

(b) Polyalphabetic Substitution (Uses more than one alphabet)

- Vigenère Cipher. (Vigenère table) -

Example:-

Encrypt → S e c r e t M e s s a g e
A p p l e A p p l e A p p

S + r c i + b + d e a v +

Decrypt → A p p l e A p p l e A p p
S + r c i + b + d e a v +
S e c r e t M e s s a g e.

(Col) }
(Row) }

↓
in A column we will see S word on which row

⑥ Steganography (hiding a secret message inside another normal looking file - image, audio)

- - Confidentiality (data remains private in 3 different situations)
 - ↓
 - At rest
 - In transit
 - In use

(AES & DES) • Symmetric Cryptosystems (used a shared secret key available to all users of cryptosystems)

(RSA) • Asymmetric Cryptosystems. (use individual combinations of public & private keys for each user of system)

- Integrity (ensures data not altered without authorization)

- Authentication (CHAP - Challenge - Handshake Authentication Protocol)

↓
A 'prove-you-are-you' test using encryption instead of sending password directly.

• Cryptographic keys

- All cryptographic algorithms rely on keys to maintain their security.

- Every algorithm has specific keyspace

- A key space is defined by its key length.

Key
↓

large
binary
number

Symmetric Key / Secret Key / Private Key Cryptography

Date: _____

- Ciphers (Algorithms used to perform encryption & decryption operations).

- Block Ciphers. (encrypts data in fixed-size chunks (blocks)).

- Stream Ciphers (operates on one character or bit of a message)

↓
Caesar Cipher
as an example.

- It relies on:
 - Plaintext
 - Keystream
 - Ciphertext

• Symmetric Key Cryptography

- Advantages →

- 1) speed ↑
- 2) 1000 to 10,000 times faster than asymmetric algorithms.
- 3) Build directly into hardware so hardware-level encryption

- Weakness → 1) Key distribution (how they will share the key safely)

- 2) No non-repudiation

↓
Bcz sender & receiver both use same key so no proof who actually sends the message.

- 3) Not Scalable → doesn't work well when there are many users.

↓

$$\text{Total Keys} = \frac{n(n-1)}{2} = \frac{100(100-1)}{2} = 4950$$

like 1 person needs to talk to person 2, 3, 4 ...

- Caesar Cipher has 26 possible keys, but only 25 useful keys (1-25). Key 0 & 26 don't change the message.

Date: _____

• Substitution Cipher

- Replaces each letter
- Change value of each

Transposition Cipher

- Rearranges order of letters.
- Change position of each

i). Caesar Cipher, formal (Algorithm)

C (Ciphertext letters)
 P (Plaintext)
 K (Key)
 E (Encryption)
 D (decryption)

$$C = E(K, P) = (P + K) \bmod 26$$

$$P = D(K, C) = (C - K) \bmod 26$$

Example:-

1) Plaintext :- HELLO

Key (K) :- 3

Formula:-

$$C = (P + K) \bmod 26$$

H $\rightarrow$ 7

$$C = (7 + 3) \bmod 26 = 10 \quad (K)$$

E $\rightarrow$ 4

$$C = (4 + 3) \bmod 26 = 7 \quad (H)$$

L $\rightarrow$ 11

$$C = (11 + 3) \bmod 26 = 14 \quad (O)$$

L $\rightarrow$ 11

(O)

O $\rightarrow$ 14

$$C = (14 + 3) \bmod 26 = 17 \quad (R)$$

(2) Cipher text : K H O O R

(3) Decryption :-

$$P (C - K) \bmod 26$$

$$K \rightarrow 10$$

$$P (10 - 3) \bmod 26 = 7 \quad (H)$$

$$H \rightarrow 7$$

$$P (7 - 3) \bmod 26 = 4 \quad (E)$$

$$O \rightarrow 14$$

$$P (14 - 3) \bmod 26 = 11 \quad (L)$$

$$O \rightarrow 14$$

$$P (14 - 3) \bmod 26 = 11 \quad (L)$$

$$R \rightarrow 17$$

$$P (17 - 3) \bmod 26 = 14 \quad (O)$$

• Brute force Attack.

↓
Try all combinations (of keys) until correct plaintext/key is found. 0-25

Best Case (1st attempt)

Worst Case (25th attempt)

Average Case (13th attempt)

• The plaintext is not always recognisable. depends upon the data type.

• How to improve Caesar Cipher?

a) Increase key space so brute force is harder

b) Change plaintext so harder to recognise structure.

Each letter in plaintext is replaced by another fixed letter. ^{Date:}

ii) Monoalphabetic Ciphers. (Simplest - Just words matching)

- Permutation (Rearranging items in every possible order)

$n!$ like $3! = 3 \times 2 \times 1 = 6$.

- In encryption, permutations are used to shuffle the data
- There are total of $26! = 4 \times 10^{26}$ possible keys making it difficult to break using brute force attacks.

iii) Playfair Cipher. (Always 5x5 matrix)

Harder for single-letter frequency but possible using digraph frequency analysis.

- Choose a keyword (e.g. Australia)
- Remove duplicates (AUSTVLi)
- Write keywords in matrix
- Treat i & j as same and same cell
- Fill remaining alphabets.

⇒ Step 1:-

a	u	s	t	v
l	i/j	b	c	d
e	f	g	h	k
m	n	o	p	q
w	x	y	y	z

⇒ Step 2:- Plain text = hello

- Before encryption, divide it into pairs of letters. If any pair contains same letter, we insert filler letter (X) between them. If odd no of letters

- Play fair $\rightarrow$ Some single plaintext letter do not always map to same ciphertext letter.
- If pair letters occurs multiple time, that pair will always encrypt to same ciphertext pair.

We also add a filler at the end:
 filler letter.

he lx lo

$\Rightarrow$ Step 3:- Encrypt pairs.

a) Pair 1:- he

$h \rightarrow (3,4)$
 $e \rightarrow (3,1)$

Both letters in same row, we shift each letter to right.

$\downarrow$
 $h \rightarrow 3,5 \rightarrow (k)$
 $e \rightarrow 3,2 \rightarrow (f)$

Encrypted pair: kf.

b) Pair 2:- lx

$l \rightarrow (3,1)$
 $x \rightarrow (5,3)$

form a rectangle, we swap Column position.

$\downarrow$
 $l \rightarrow (2,3) \rightarrow b$
 $x \rightarrow (5,1) \rightarrow v$

c) lo

$l \rightarrow (2,1)$
 $o \rightarrow (4,3)$

$\downarrow$
 bm.

kf bv bm.

iv) ^(OTP) One-time Pad (Algorithm)
↓

The key must be completely random as long as the plaintext, and never reused.

- Only known unbreakable cipher
 - Ciphertext has no statistical relationship with plaintext
- Limitations →
 - Difficult to create large number of random keys
 - Distributing unique long random keys is difficult.

Transportation Techniques:-

v) 9) Rail Fence Cipher Encryption.

- a) Write message in zigzag pattern across fixed number of rows
- b) Number of rows = your key (depth)
- c) Read row by row to get Cipher-text.

Example:-

Plaintext :- security and cryptography

key:- 4

1 s t r r
2 e i y e y g a
3 c r a d p o p y
4 u n t h y

Cipher text :- S t r e i g c y g s c r a d
P o p y u n t h.

b) Rows & Column Ciphers.

- We decide how many col we use
- Write message row by row in a grid
- read columns by column in order of key numbers.

Example:-

plaintext :- Security and cryptography

key :- 315624

- We choose 6 col as 6 digits and write it

1	2	3	4	5	6
S	e	c	u	r	i
t	y	a	n	d	c
r	y	p	t	o	g
r	a	p	h	y	x

- Now read cols as 315624.

3	c	a	p	p
1	s	t	r	r
5	r	d	o	y
6	i	c	g	x
2	e	y	y	s
4	u	n	t	h

- Cipher text = cappsterrdayicgxeysunth.